KL UNIVERSITY
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CAPSTONE PROJECT

Music Playlist & Listening Stats Manager



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Abstract



The Music Playlist and Listening-Stats Manager is a simple Java-based application used to manage songs and track listening habits. It lets users add songs, store and display song details, and record how many times each song is played, making it easy to organise music and see which songs are played the most.

Song search, favourites, sorting, multiple playlists, and a graphical interface can be added.

Introduction

What It Is
A simple Java-based application to manage songs and track listening habits.

What It Does
Adds songs, stores song details, displays your music collection, and counts plays per song.

How It Works
Built using arrays, loops, conditions, switch statements, operators, and methods.

Where It Goes
Future scope includes song search, sorting, multiple playlists, favourites, and a GUI.



Course Outcomes: CO1–CO3



CO1

Data Types & Operators

Data types and operators store song details such as name, duration, and play count, and calculate listening statistics.



CO2

Control Structures

Conditions, loops, switch statements, and break control program flow, handle user choices, and run the interactive menu.



CO3

Methods & Arrays

Methods organise operations like adding and displaying songs, while arrays store multiple song details in memory.

Course Outcomes: CO4–CO6



CO4

OOP & Exceptions

Song and Playlist classes with encapsulated fields, inheritance for song types, and try-catch for invalid input.



CO5

Strings & File I/O

String methods search by title or artist, Comparator sorts by play count, and files save playlists.



CO6

Collections & Streams

ArrayList and HashMap store songs and play counts, with Streams to filter, sort, and group listening stats.

Project Objectives



PRIMARY GOAL

Develop a user-friendly platform for playlist management and music analytics that is accessible, organised, and insightful.

CORE FOCUS AREAS

1. Playlist creation and custom organisation
2. Real-time listening statistics tracking
3. Music consumption pattern analysis
4. Secure authentication and data persistence
5. Music search and discovery functionality

Advantages

User-Centric Design

Intuitive interface makes playlist management effortless for all users.

Real-Time Analytics

Instant insights into listening habits, preferences, and play counts.

Data Visualization

Charts and graphs make statistics easy to interpret at a glance.

Security

Secure authentication protects user data and privacy effectively.

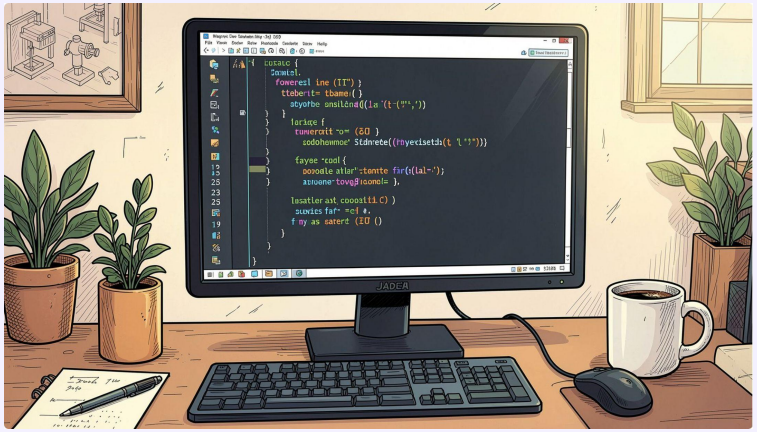
Accessibility

Web-based platform accessible across all devices and browsers.

Personalization

Customizable playlists and preference tracking for every user.

Project Development



Built Entirely in Java

Developed using core Java concepts taught in the course, without any external libraries or frameworks.

1 Arrays

Used to store song names, durations, and play counts.

2 Methods

Separate methods add, display, and update songs.

3 Loops & Conditions

Control program flow and handle user input validation.

4 Switch Statement

Powers the interactive menu for navigating features.

Implementation & Demo

A walkthrough of the core features of the Music Playlist & Listening Stats Manager

1	**User Registration & Login**	Secure registration and login to protect personal data.
2	**Playlist Management**	Create, edit, and delete playlists, and add or remove songs with ease.
3	**Listening Statistics Dashboard**	Real-time dashboard tracking play counts, top songs, and listening trends.
4	**Search & Filter**	Search and filter music by genre, artist, or title for quick discovery.
5	**CRUD Operations**	Full Create, Read, Update, and Delete operations on playlists and user data.

Conclusion



Our Music Playlist and Listening Stats Manager gives a simple, organised way to manage songs and track listening activity.

- Applied arrays, loops, conditions, switch statements, and methods to a real-world application.
- Strengthened our Java skills and showed how programming solves real problems.
- Ready to grow with song search, sorting, multiple playlists, favourites, and a GUI.

Future Scope

AI Recommendations

Machine learning to suggest songs based on listening patterns.

Social Features

Share playlists, collaborate with friends, and discover trending music.

Mobile App

Native iOS and Android apps for an enhanced mobile experience.

Streaming Integration

Direct integration with Spotify, Apple Music, and YouTube Music.

Offline Mode & Gamification

Download playlists and earn achievements, badges, and leaderboards.

Thank you

